

Transforming Global Database Operations through 24x7 Multi-Platform DBA Support

Introduction

A leading U.S.-based global asset management firm with an AUM of \$740 billion sought a trusted technology partner to manage its end-to-end database operations across multiple geographies. The firm provides investment management and research services to institutional, high-net-worth, and retail investors across the U.S., Europe, Asia, and Australia.

With a complex mix of on-premises and cloud-based environments, spanning multiple RDBMS platforms; including Microsoft SQL Server, Oracle, SAP ASE (Sybase), SAP IQ (SybaseIQ), and PostgreSQL, the firm needed a partner capable of ensuring 24x7 operational continuity, scalability, and compliance across more than 600 database servers and 100+ business-critical applications.

NuWare, leveraging its extensive experience in Database Administration (DBA) and Engineering Services, was selected to provide L1, L2, and L3 DBA support under a flexible offshore model designed to meet stringent SLAs and evolving business needs.

The Challenge

The firm's global IT infrastructure presented a range of operational and governance challenges that required specialized expertise and constant vigilance:

1. Multi-Platform Database Environment

The client operated on a hybrid mix of on-premises and cloud databases across multiple RDBMS technologies, increasing complexity in maintenance and patch management.

2. High Volume of Alerts and Requests

Managing over 4,000 tickets per month, the support team faced continuous pressure to maintain SLAs with rapid response and resolution times.

3. Global Datacenter Footprint

Database servers spanned U.S., Europe, Asia, and Australia, requiring continuous 24x7 support coverage and effective time zone coordination.

4. VLDB and Mission-Critical Systems

Handling Very Large Databases (VLDBs) supporting trading, research, and portfolio systems demanded high performance, data integrity, and rapid recovery capabilities.

5. **Continuous Patching and Compliance**

Maintaining software versions, applying security patches, and performing database health checks were essential to ensure regulatory compliance and data security.

6. **Complex Service Mix and High Availability**

Balancing routine DBA requests, performance tuning, and disaster recovery testing across both on-premise and cloud setups required strong automation and governance.

The client needed a resilient, proactive, and automation-enabled support model to ensure continuous database uptime, quick response to incidents, and seamless collaboration with global development teams.

NuWare's Approach

NuWare established a multi-tiered Database Support and Engineering framework that combined 24x7 offshore L1 monitoring, L2 business-hour application DBA support, and L3 database engineering services.

The solution was structured to deliver flexibility, automation, and proactive performance optimization across all database environments.

1. 24x7 L1 DBA Support

NuWare's L1 team provided continuous, SLA-driven monitoring and first-level response for all production and non-production databases.

Key functions included:

- **Server alerts and warning management:** 24x7 proactive monitoring of server health and thresholds.
- **Database builds, patching, and migrations:** Automated provisioning and version upgrades across SQL, Oracle, and Sybase environments.
- **Disaster Recovery (DR) support:** Scheduled DR testing and failover validations.
- **Health checks and maintenance:** Routine performance checks and index optimization.
- **User service requests:** Security management, DB object creation/modification, and code deployments (SSIS/SSRS).
- **Change management:** Controlled patch and configuration rollouts.
- **Performance tuning:** Real-time monitoring of critical batch jobs and query optimization.

- **Documentation and SOP management:** Continuous updates to runbooks and process documentation.

Average Monthly Volume of L1 Tasks:

- Server alerts and warnings: 1,400
- Change requests: 650
- User requests: 600
- Database refreshes: 150
- Script/file deployments: 250
- Other DBA services: 600

2. L2 Application DBA Support

The L2 team delivered advanced support and collaboration with application development groups globally.

Key activities included:

- **Requirement analysis and solutioning:** Partnering with application managers to design optimal DB configurations.
- **Proactive health checks:** Identifying and addressing bottlenecks before impacting performance.
- **Capacity and cost optimization:** Upsizing or downsizing servers based on utilization metrics.
- **On-call performance management:** Rapid diagnosis and resolution of performance degradation and outages.

3. L3 Database Engineering

NuWare's Database Engineering team acted as subject matter experts (SMEs), driving automation, tooling, and continuous improvement.

Major contributions included:

- **DBA Task Automation:** Automated recurring tasks like database refreshes, DR failovers, and sanity checks.
- **Custom Maintenance Scripts:** Developed bespoke scripts for backup, log shipping, and replication monitoring.
- **Advanced Monitoring Scripts:** Created SQL and PowerShell scripts to track blocking, deadlocks, long-running transactions, and disk utilization.
- **Server and Audit Reports:** Automated daily health check and latency reports for DBAs and compliance teams.

- **Certification and Tool Validation:** Conducted evaluations of new DB tools, technologies, and cloud platforms before rollout.

Outcomes

The partnership resulted in measurable improvements across reliability, scalability, and cost efficiency.

1. Uninterrupted Global Operations

- Achieved 99.9% database uptime across all RDBMS platforms.
- Supported real-time data synchronization between on-premise and cloud systems.

2. SLA Compliance and Incident Response

- Maintained **<10-minute acknowledgment** for P1 alerts and **under 30-minute resolution** for critical incidents.

3. Operational Efficiency and Cost Reduction

- Automated repetitive DBA tasks, resulting in **35% reduction in manual workload**.
- Optimized server utilization, reducing infrastructure cost by **20% annually**.

4. Enhanced Performance and Security

- Improved query performance across major trading databases by **25%** through tuning and index optimization.
- Strengthened compliance with **regular patching, DR drills, and audit reporting**.

5. Global Scalability and Flexibility

- Successfully supported over **600 database servers** across hybrid environments (Azure, AWS, on-prem).
- Delivered consistent service across **multiple time zones and business units**.

Future Outlook

Building on its success, NuWare continues to evolve the DBA support model to align with emerging trends in cloud and automation:

- **AI-Driven Performance Monitoring:** Integrating anomaly detection to preempt potential failures.
- **Self-Healing Database Automation:** Enabling automated recovery for resource bottlenecks and deadlocks.
- **Cloud-Native Database Management:** Expanding support for fully managed services like Azure SQL and AWS RDS.
- **DataOps Integration:** Streamlining database operations into CI/CD pipelines for faster delivery.

These initiatives aim to create a predictive, intelligent, and autonomous DBA ecosystem, ensuring resilience and agility across the client's global operations.